

Code Converter

It is a combinational circuit which converts one type code into another type of code.

Ques. Design a code converter which converts 4-bit binary code into grey code.

b_3	b_2	b_1	b_0	g_3	g_2	g_1	g_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

from truth table,

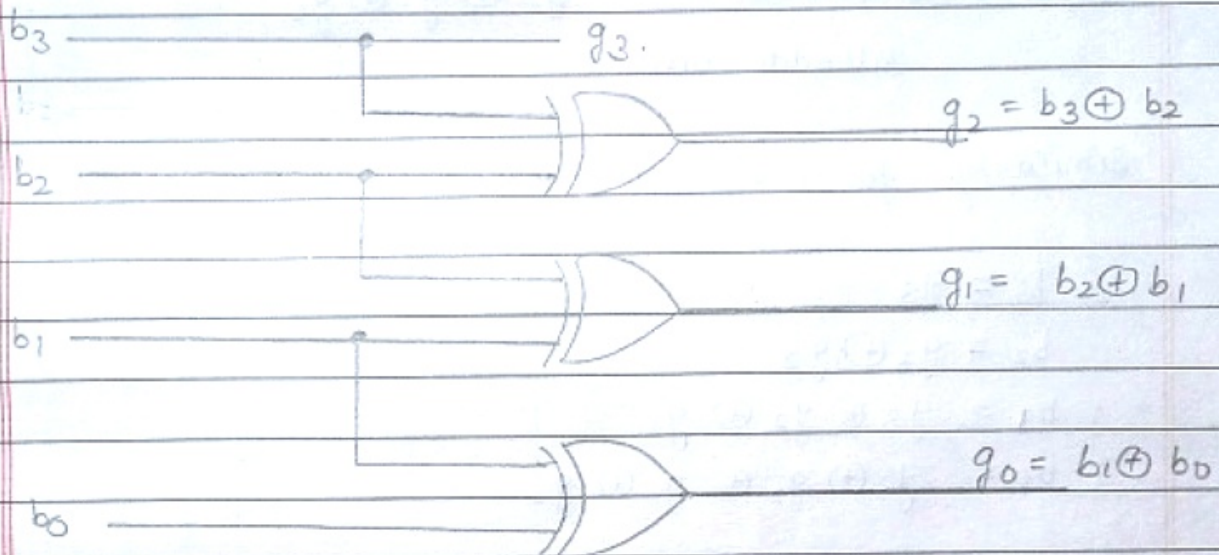
$$g_3 = \sum m(8, 9, 10, 11, 12, 13, 14, 15) = b_3$$

$$g_2 = \sum m(4, 5, 6, 7, 8, 9, 10, 11) \quad (\because \text{It's XOR of } b_3 \text{ \& } b_2)$$

$$= \overline{b_3}b_2 + b_3\overline{b_2}$$

$$g_1 = b_2 \oplus b_1$$

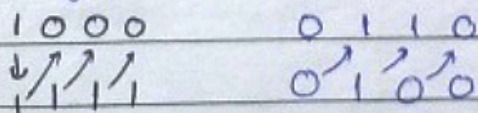
$$g_0 = b_1 \oplus b_0$$



Ques. Design a code converter which converts 3-bit gray code into binary code.

g_2	g_1	g_0	b_2	b_1	b_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	1	1	0
1	1	0	1	0	0
1	1	1	1	0	1

⊛ diagonally



$$b_2 = \sum m(4, 5, 6, 7) = g_2$$

$$b_1 = \sum m(2, 3, 4, 5) = g_2 \oplus g_1$$

$$b_0 = \sum m(1, 2, 4, 7) = g_2 \oplus g_1 \oplus g_0$$

↓
full adder sum

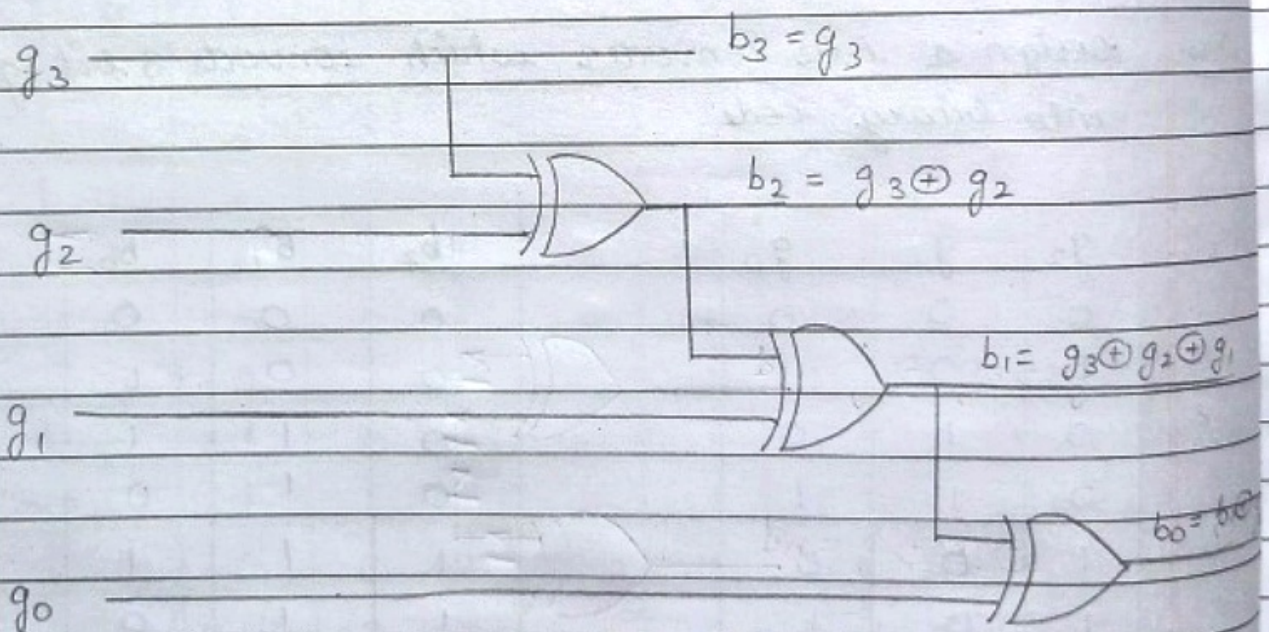
Similarly, for 4-bit conversion

$$b_3 = g_3$$

$$b_2 = g_3 \oplus g_2$$

$$b_1 = g_3 \oplus g_2 \oplus g_1$$

$$b_0 = g_3 \oplus g_2 \oplus g_1 \oplus g_0$$



BCD to XS-3

3 to
add BCD

8	4	2	1					
B_3	B_2	B_1	B_0		X_3	X_2	X_1	X_0
0	0	0	0		0	0	1	1
0	0	0	1		0	1	0	0
0	0	1	0		0	1	0	1
0	0	1	1		0	1	1	0
0	1	0	0		0	1	1	1
0	1	0	1		1	0	0	0
0	1	1	0		1	0	0	1
0	1	1	1		1	0	1	0
1	0	0	0		1	0	1	1
1	0	0	1		1	1	0	0

$$BCD = \sum d(10, 11, 12, 13, 14, 15)$$

$$X_3 = \sum m(5, 6, 7, 8, 9) + \sum d(10-15)$$

$$X_2 = \sum m(1, 2, 3, 4, 9) + \sum d(10-15)$$

$$X_1 = \sum m(0, 3, 4, 7, 8) + \sum d(10-15)$$

$$X_0 = \sum m(0, 2, 4, 6, 8) + \sum d(10-15)$$

$B_3 B_2$	$B_1 B_0$	00	01	11	10
00					
01			1	1	1
11		x	x	x	x
10		1	1	x	x

$$X_3 = B_3 + B_2 B_0 + B_2 B_1$$

X_2				
1	1	1	1	
x	x	x	x	
1	x	x	x	

$$X_2 = B_3' B_1 + B_2 B_0 + B_2 B_1 B_3'$$

X_1				
1		1		
1		1		
x	x	x	x	
1		x	x	

$$X_1 = B_3 \oplus B_0$$

X_0				
1				1
1				1
x	x	x	x	x
1		x	x	x

$$X_0 = B_0'$$

XS-3 to BCD.

(3-12) valid.

0, 1, 2, 12-15 $\rightarrow$ invalid.

B	4	2	1		B ₃	B ₂	B ₁	B ₀
X ₃	X ₂	X ₁	X ₀					
0	0	0	0		0	0	0	0
0	0	1	0		0	0	0	1
0	1	0	0		0	0	1	0
0	1	0	1		0	0	1	1
0	1	1	0		0	1	0	0
0	1	1	1		0	1	0	1
1	0	0	0		0	1	1	0
1	0	0	1		0	1	1	1
1	0	1	0		1	0	0	0
1	0	1	1		1	0	0	1
1	1	0	0		1	0	0	1

$$\Sigma d = \Sigma (0, 1, 2, 12, 13, 14, 15)$$

$$B_3 = \Sigma m(8, 9) + \Sigma d$$

$$B_2 = \Sigma m(4, 5, 6, 7) + \Sigma d$$

$$B_1 = \Sigma m(2, 3, 6, 7) + \Sigma d$$

$$B_0 = \Sigma m(1, 3, 5, 7, 9) + \Sigma d$$